

Vector Angle Calculation

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Question

If $|\mathbf{a}| = 2$, $|\mathbf{b}| = 7$ and $\mathbf{a} \times \mathbf{b} = 3\hat{i} + 2\hat{j} + 6\hat{k}$, find the angle between $\mathbf{a}$ and $\mathbf{b}$.

Theoretical Formula

The magnitude of the cross product of two vectors is defined by the formula:

$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin(\theta) \quad (1)$$

To use this, we first need to calculate the magnitude of the given cross product vector.

Solution using Matrices

Let the vector $\mathbf{c} = \mathbf{a} \times \mathbf{b}$. We represent it as a column matrix:

$$\mathbf{c} = \begin{pmatrix} 3 \\ 2 \\ 6 \end{pmatrix} \quad (2)$$

The squared magnitude is found by $\mathbf{c}^T \mathbf{c}$:

$$\begin{aligned} \|\mathbf{c}\|^2 &= \begin{pmatrix} 3 & 2 & 6 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \\ 6 \end{pmatrix} \\ &= (3 \cdot 3) + (2 \cdot 2) + (6 \cdot 6) = 9 + 4 + 36 = 49 \end{aligned}$$

So, $|\mathbf{c}| = \sqrt{49} = 7$.

Finding the Angle

Now, substituting the values into Equation (??):

$$\begin{aligned} |\mathbf{a} \times \mathbf{b}| &= |\mathbf{a}||\mathbf{b}| \sin(\theta) \\ 7 &= (2)(7) \sin(\theta) \\ 7 &= 14 \sin(\theta) \end{aligned}$$

Solving for $\sin(\theta)$:

$$\sin(\theta) = \frac{7}{14} = \frac{1}{2}$$

Therefore, the angle is $\theta = \arcsin\left(\frac{1}{2}\right) = 30^\circ$ or $\frac{\pi}{6}$ radians.

Solution 1: Pure C Code

```
#include <stdio.h>
#include <math.h>

#ifndef M_PI
#define M_PI 3.14159265358979323846
#endif

double angle_between(double a, double b, double x, double y,
    double z) {
    double cross_mag = sqrt(x*x + y*y + z*z);
    double sin_theta = cross_mag / (a * b);
    double theta_rad = asin(sin_theta);
    return theta_rad * 180.0 / M_PI;
}

int main() {
    double angle = angle_between(2.0, 7.0, 3.0, 2.0, 6.0);
    printf("Angle between a and b = %.2f degrees\n", angle);
    return 0;
}
```

Solution 2: Python + C (The C Library)

```
#include <math.h>

#ifndef M_PI
#define M_PI 3.14159265358979323846
#endif

double angle_between(double a, double b, double x, double y,
    double z) {
    double cross_mag = sqrt(x*x + y*y + z*z);
    double sin_theta = cross_mag / (a * b);
    double theta_rad = asin(sin_theta);
    return theta_rad * 180.0 / M_PI;
}
```

Solution 2: Python + C (The Python Script)

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt
import math

lib = ctypes.CDLL("./vector.so")
lib.angle_between.argtypes = [ctypes.c_double, ctypes.c_double,
                               ctypes.c_double, ctypes.c_double,
                               ctypes.c_double]
lib.angle_between.restype = ctypes.c_double

a_mag = 2.0
b_mag = 7.0
cross = np.array([3.0, 2.0, 6.0])

theta = lib.angle_between(a_mag, b_mag, cross[0], cross[1], cross
                          [2])
```


Solution 2: Python + C (The Python Script)

```
a = np.array([a_mag, 0, 0])
b = np.array([b_mag*math.cos(math.radians(theta)),
              b_mag*math.sin(math.radians(theta)), 0])

fig = plt.figure(figsize=(8,6))
ax = fig.add_subplot(111, projection='3d')
ax.quiver(0,0,0, *a, color='r', label='a (|a|=2)')
ax.quiver(0,0,0, *b, color='b', label='b (|b|=7)')
ax.quiver(0,0,0, *cross, color='g', label='a x b')
ax.legend()
plt.show()
```

Solution 3: Pure Python

```
import math
import numpy as np
import matplotlib.pyplot as plt

a_norm = 2.0
b_norm = 7.0
cross_ab = np.array([3.0, 2.0, 6.0])

cross_norm = np.linalg.norm(cross_ab)
dot_sq = (a_norm**2) * (b_norm**2) - cross_norm**2
dot_val = math.sqrt(dot_sq)

cos_theta = dot_val / (a_norm * b_norm)
theta_rad = math.acos(cos_theta)
theta_deg = math.degrees(theta_rad)
```

Solution 3: Pure Python

```
fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')
ax.quiver(0, 0, 0, a_norm, 0, 0, color='r', label='a')
ax.quiver(0, 0, 0, 0, b_norm, 0, color='g', label='b')
ax.quiver(0, 0, 0, cross_ab[0], cross_ab[1], cross_ab[2], color='b', label='a x b')
ax.legend()
plt.show()
```

Plot Result

